

Abstract algebra – 6

The hardest problems

Homework

13 сентября 2026 года

Theoretical minimum. Nothing new: the tools are those of the first five sheets. Count the same quantity twice (class equation, orbit–stabiliser, the pairing $x \leftrightarrow x^{-1}$); reduce a ring to a field (a finite ring without zero divisors is a field; quotient by a maximal ideal); linearise the group by passing to functions on it (convolution, the averaging operator, and for abelian groups the Fourier transform over the character group). Lagrange, Cauchy, the class equation, the classification of finite abelian groups, Wedderburn, Jacobson, Maschke, Schur, the simplicity of A_n for $n \geq 5$ and Sylow’s theorems may be quoted; everything else should be proved. The three traps are non-commutativity, characteristic two, and infinity, where pigeonhole and averaging both fail.

1. For a prime p put $G_p = \bigcup_{n \in \mathbb{N}} \{z \in \mathbb{C} : z^{p^n} = 1\}$. Prove that G_p is a subgroup of $(\mathbb{C}^*, \cdot)$, and that an infinite subgroup $H \leq (\mathbb{C}^*, \cdot)$ has all of its proper subgroups finite if and only if $H = G_p$ for some prime p . (Румыния, 1998)
2. For a finite additive group L let $i(L)$ be the number of multiplicative operations turning it into a ring. Prove that (a) $i(\mathbb{Z}_{12}) = 4$; (b) $i(A \times B) \geq i(A) i(B)$ for finite commutative A, B ; (c) there are sequences of finite commutative groups G_k, H_k with $|G_k|/i(G_k) \rightarrow 0$ and $|H_k|/i(H_k) \rightarrow \infty$. (Румыния, 2003)
3. Let $f: \mathbb{N} \rightarrow \mathbb{Z}$ be a non-constant function such that for every finite group G and all $g, h \in G$

$$f(o(g^{o(h)} h^{f(o(g))})) = f(o(h^{o(g)} g^{f(o(h))})),$$

where $o(a)$ denotes the order of a . Prove that $f(1)$ is odd. (IMC 2025, shortlist)

4. Show that in a finite division ring with at least four elements the set of non-zero elements can be partitioned into two non-empty sets A and B with $\sum_{x \in A} x = \prod_{y \in B} y$. (Румыния, 2015)
5. Let $(R, +, \cdot)$ be a ring and let f be a surjective endomorphism of R with $[x, f(x)] = 0$ for all x , where $[a, b] = ab - ba$. Prove that (a) $[x, f(y)] = [f(x), y]$ and $x[x, y] = f(x)[x, y]$ for all x, y ; (b) if R is a division ring and f is not the identity, then R is commutative. (Румыния, 2012)
6. Say that a ring $(A, +, \cdot)$ has property (P) if $|A| \geq 4$, the element $1 + 1$ is invertible, and $x + x^4 = x^2 + x^3$ for all $x \in A$. Prove that if A has property (P) and $a, b \in A$ are distinct elements with a and $a + b$ units, then $1 + ab$ is a unit but b is not; and give an example of a ring with property (P). (Румыния, 2025)
7. Call a finite group $(G, \cdot)$ of order $n \geq 2$ *arrangeable* if its elements can be ordered so that $G = \{a_1, \dots, a_n\} = \{a_1 a_2, a_2 a_3, \dots, a_n a_1\}$. Determine all n for which $(\mathbb{Z}_n, +)$ is arrangeable, and give an example of an arrangeable group of even order. (Румыния, 2023)
8. Let p be a prime and let W be the smallest set of polynomials over $\mathbb{F}_p$ such that $x + 1$ and $x^{p-2} + x^{p-3} + \dots + x^2 + 2x + 1$ belong to W , and for any $h_1, h_2 \in W$ the remainder of $h_1(h_2(x))$ modulo $x^p - x$ also belongs to W . How many polynomials does W contain? (IMC)
9. Let p be a prime. Call n *interesting* if $x^n - 1 = (x^p - x + 1)f(x) + pg(x)$ for some polynomials f, g with integer coefficients. Prove that $p^p - 1$ is interesting, and determine for which p it is the smallest interesting number. (IMC)
10. Let A be a ring and let D be the set of its non-invertible elements. Assume $a^2 = 0$ for every $a \in D$. Prove that (a) $axa = 0$ for all $a \in D$ and all $x \in A$; (b) if D is finite with at least two elements, then there is a non-zero $a \in D$ with $ab = ba = 0$ for every $b \in D$. (Румыния, 2016)